

Sectors in the sky

What “the sector is the cause” says about dark matter, dark energy and K3 – three machine-checked lattice identities, two discriminating questions, and the routes it closes

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SocrateAI-Scientific-QuantumFluids

<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

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Abstract

A programme on two-dimensional quantum fluids ended with a reading it could defend: a duality is a reindexing of a lattice of topological sectors that preserves a spectrum, and the physics happens when a sector crosses a threshold, proliferates, annihilates, or is imposed [1]. This note asks what the reading says when the U(1) phase field is a dark-matter halo, when the integer sector is a four-form flux, and when the lattice is $H^2(K3, \mathbb{Z})$. It is written under a transfer rule stated on the first page: the chain *integrality* $\rightarrow$ *conservation* $\rightarrow$ *barrier* $\rightarrow$ *scale additivity* $\rightarrow$ *continuum limit* transfers to any U(1) field and so does the method of the matched control arm; the coupling between the integer and the observables does not, so no number crosses from the torus to the sky. Within that rule we prove in Lean 4, against Mathlib and under two independent kernels, fifteen elementary statements (**ChargeLattice.lean**): Moore’s dyon discriminant on the $K3 \times T^2$ charge lattice is $SL(2, \mathbb{Z})$ -invariant, $D(ap + bq, cp + dq) = (ad - bc)^2 D(p, q)$, so the horizon area reads a level set the duality merely permutes; on the Bousso–Polchinski flux lattice one nucleation changes Λ by exactly $-(n_i - \frac{1}{2})q_i^2$ and lowers it iff $n_i \geq 1$; Kaloper’s discretely evanescent dark energy vanishes iff its top-form sector is the CP-invariant one and the discharge from $N \leq 1$ ends after exactly $1 - N$ steps. From these and the literature we draw two discriminating questions – for wave dark matter, whether a claimed vortex signature survives setting the halo’s net circulation to zero at fixed energy (sector versus interference pairs); for dark energy, whether a $w(z) \neq -1$, if it survives, is a continuous roll or a discrete cascade – and we record what the reading closes: the attractor mechanism makes the K3 an *output* of the charges, so the fluid $\leftrightarrow$ K3 identification this programme excluded could never have been an input; and topological-defect dark energy is excluded by data as the dominant component. Nothing here is new physics; the mathematics is bilinearity and a square; what is new is that the distinction between the lattice and the event on it is stated so that a kernel can check it and a survey can test it.

Transfer rule (in force throughout). From the quantum-fluid laboratory [2, 3] the following transfers to any U(1) phase field: the integer sector, its conservation by every continuous evolution without a phase slip, the barrier a slip must cross, the scale additivity $W(R) = \text{net enclosed charge}$, and the sampled-to-continuum reading of the loop sum. The *method* transfers: a causal claim about a label needs a control arm that is equally arbitrary, and the label must be read at a stated scale. What does *not* transfer is the coupling between the integer and the observables – in the fluid, η , $n_s \lambda_T^2$, the condensate drop, T_v/T_b – which is empirical and specific. This note therefore carries structure and method, never a number.

1 The reading, in one paragraph

In every case examined in [1] – compact boson, Ising, string gas, superconductor–insulator, quantum Hall – the same triad appeared: a group of sectors G ($\mathbb{Z}$ windings, $\mathbb{Z}_2$ order, Chern numbers, string windings); a duality that is a Fourier or reindexing map between G -sectors and $\hat{G}$ -sectors preserving the spectrum [4]; and physics that happens at an *event on the lattice*. For cosmology the questions are then fixed in advance: what is the sector group, what function of the sector does the observable read, what is the event, and what is the duality doing besides relabelling?

2 Dark matter as a U(1) phase field: sector versus pairs

The sector. Bosonic dark matter lighter than ~ 30 eV is wave-like in a Milky-Way halo [5, 6]: a field $\psi = \sqrt{\rho} e^{iS}$ whose U(1) is particle number; the winding of S around a loop is an integer and the circulation is quantised in units of $2\pi/m$. This is the **VortexWinding** setting with another m and no box.

What is established. Hui, Joyce, Landry and Li [7] show that interference inside a virialised halo *inevitably* produces vortices: rings in three dimensions, about one per de Broglie volume, of winding ± 1 only, with $\rho \propto r_\perp^2$ and $v \propto 1/r_\perp$ near the line, moving at $\sim 1/(m\ell)$, reconnecting rather than frustrating, and present even in a halo of zero net angular momentum; their lensing imprint is a 5–10% flux anomaly in strongly lensed systems. Zhou, Leung, Poon and Chu [8] show that a *rotating* condensate halo forms a vortex lattice of under-density columns whose lensing signature is a regularly spaced brightness anomaly, visible at $\lesssim 10$ mas resolution when the rotation axis is along the line of sight. Berezhiani and Khoury [9] estimate $\sim 10^{23}$ vortices of millimetre core in a superfluid halo, individually invisible. Zhou et al. pose our exact question themselves, unanswered: “it will be important for a future study to identify ways to distinguish if the magnification anomalies are caused by underdensity ‘holes’ (due to vortices) or overdensity subhalos”; every one of their simulated haloes has $\Omega > 0$ built into the initial condition, none is run at $\Omega = 0$ with a compensated vortex population at matched energy. Brax and Valageas [10], in independent 3D simulations of a rotating self-interacting halo, find the same two populations we do – a regular lattice aligned with the total angular momentum inside the relaxed core (the sector) and “a tangle of intertwining and curved vortex lines of any direction” with random ± 1 windings in the envelope (the pairs) – and flag the same open problem: distinguishing the two is “a difficulty for numerical computations.” Rindler-Daller and Shapiro [11] add a second, independent condition: *a sector vortex additionally requires strong-enough self-interaction* (their threshold is $\lambda_{\text{dB}} \lesssim R$); in the pure-quantum-pressure regime Hui et al. and Zhou et al.’s interference vortices study, nonzero angular momentum is not sufficient by itself. The vortex density itself, $\pi/\lambda_{\text{dB}}^2$, is not special to dark matter: it is Berry and Dennis’s 25-year-old result for any isotropic random wave field [12], and the perimeter-law screening of its net charge is Dennis’s [13].

What the sector reading adds. Three statements, one prediction, no number.

1. *Rings are pairs.* By scale additivity (**ScaleResolvedWinding**: $W(R) = \text{net enclosed charge}$), a vortex ring pierces any plane at two points of opposite winding and is invisible to every sector observable read at $R \gg \lambda_{\text{dB}}$. Hui et al.’s rings are the halo’s thermal pairs – they carry energy, scatter and heat, exactly as the pairs of our round 2 cooled the bath and their annihilation reheated it – and they are not the halo’s sector.
2. *The only sector-level observable of a halo is its net circulation.* The sum of the windings of all lines threading a large loop is the quantised circulation, i.e. the angular momentum in units of $\hbar$ per particle. Zhou et al.’s lattice *is* the sector (its vortex count is set by Ω); Hui et al.’s

rings are the pairs. One measurement separates them: W at large R . This is the distinction between our persistent current (winding -3 , intact after 1500 time units, 29.5% of the atoms at $k = (-3, 0)$) and our coarsening arm ($W = 0$).

3. *The barrier is low.* A halo’s winding changes when a line crosses the boundary or nucleates at a density zero (`TopologicalProtection`); a halo has no wall, its “boundary” is where $\rho \rightarrow 0$, and lines leave freely. The sector is real, but it is not protected the way a ring current is.

Prediction (falsifiable, unfitted). In a rotating wave-dark-matter halo simulation, the lattice vortices’ lensing signature scales with the imposed net circulation W at fixed total energy; the ring vortices’ signature does not change when W is set to zero at the same energy. A “vortex lensing anomaly” that survives $W = 0$ is interference substructure, not a sector effect, and a claim of “condensate dark matter seen through its vortices” resting on it is a claim about pairs. This is the round-2 protocol (topology arm against an energy-matched control arm) written for a halo; the instrument is $W(R)$ read from the phase. On our own periodic torus this holds without qualification; on a bounded ring or a real halo boundary, a pair with one member crossing the boundary is the literal microscopic mechanism by which net winding changes by one unit [14] – the “pairs are invisible” claim needs that boundary-leak caveat off the torus. No published simulation has yet run the $\Omega = 0$ control this prediction calls for [10].

3 Dark energy as a sector lattice: one algebra, four models, the honest limit

Bousso–Polchinski. With J four-form fluxes of charges q_i and integer quanta n_i [15], $\Lambda(n) = \Lambda_{\text{bare}} + \frac{1}{2} \sum_i n_i^2 q_i^2$ is a function on $\mathbb{Z}^J$; membrane nucleation (the Brown–Teitelboim process [16]) moves one step and lowers Λ ; with $J \sim 100$ the discretuum near zero is dense enough. What selects the observed value is anthropic in the weakest sense.

Theorem 1 (`bp_step`, `bp_step_neg_iff`, `bp_ge_bare`, `bp_min_at_zero`). $\Lambda(n - e_i) - \Lambda(n) = -(n_i - \frac{1}{2})q_i^2$; the step lowers Λ iff $n_i \geq 1$; $\Lambda(n) \geq \Lambda(0)$ with equality iff $n = 0$ when every $q_i \neq 0$.

Kaloper. A hidden $SU(N)$ confining at $\Lambda_{\text{dark}} \sim 10^{-3}$ eV [17]: the top-form flux is quantised, $\theta_{\text{dark}} = (1 - N)\hat{\theta}$, $V = \frac{1}{2}X\theta_{\text{dark}}^2$; millimetre bubbles nucleate at $\Gamma \sim H_0^4$ and lower V in discrete random steps to exactly zero at the CP-invariant sector; $w \neq -1$ transiently; acceleration may cease within $\sim 1/H_0$; $\Omega_{\text{GW}} < 10^{-9}$ at $f \lesssim 10^{-12}$ Hz.

Theorem 2 (`kaloper_step`, `kaloper_min`, `kaloper_terminates`). $V(N) - V(N + 1) = X\hat{\theta}^2((1 - N) - \frac{1}{2})$; $V \geq 0$ with $V = 0$ iff $N = 1$; from $N \leq 1$ the discharge reaches $V = 0$ after exactly $1 - N$ nucleations.

This is the phase-slip cascade of our $e = 0.60$ arm – winding $2 \rightarrow 0$ by slips, the current’s energy going to the bath – written for the vacuum. The model’s content is the *rate*; the algebra is the sector’s.

QCD sectors. Van Waerbeke and Zhitnitsky [18] derive dark energy from tunnelling between QCD θ -sectors in an expanding background, $\rho_{\text{DE}} \approx c_H \Lambda_{\text{QCD}}^3 H$, with no new field and a $w(z)$ that may cross -1 . The linear-in- H leftover is proved on $H^3 \times S^1$ and conjectured for de Sitter.

Topological-defect dark energy: excluded as the dominant component. Frustrated non-Abelian string networks give $w = -1/3$, domain-wall networks $w = -2/3$, stabilised by a shear modulus [19, 20, 21]; here the sector is literal ($\pi_1(G/H)$). Current data exclude a string-network component above $\Omega_s \gtrsim 1\%$: $\Omega_s < 0.0082$ at 95% from CMB+DESI, with the Bayesian evidence favouring Λ CDM in every variant [22]. A few-percent domain-wall component ($\Omega_{\text{dw}} \approx 5\%$) is not excluded the same way – it is mildly preferred with one supernova sample (DESY5) and disfavoured with the others [23] – so the honest statement is narrower than “excluded”: subdominant, dataset-dependent, and not established.

Discriminating question, now with an available test. In all four models the vacuum energy is a function of an integer sector, the dynamics a walk on the lattice by slips, the end point a distinguished sector. The question that separates the lattice from no lattice: is a $w(z) \neq -1$, if it survives, a *continuous roll* (quintessence: no integer) or a *discrete cascade* (steps in ρ_{DE} , a finite number of nucleated bubbles per Hubble volume)? DESI DR2’s own binned and Gaussian-process reconstructions of $w(z)$ are consistent with the smooth $w_0 w_a$ CPL fit, with no preference for extra freedom [24, 25]; the background expansion history alone cannot separate a roll from a cascade at current precision. It can be separated: a completed first-order transition releasing a fraction r of the dark energy leaves a CMB-anisotropy signature – fluctuations from the spatially random local completion time, with no analogue in a homogeneous roll – bounded by $r \lesssim 10^{-5}(\beta/H_\star)^2$ [26], four to five orders of magnitude tighter than the background-level bound ($r \lesssim 0.65$ from BAO alone). A genuinely discrete cascade model already fits DESI DR2 comparably to CPL with $O(1\text{--}10)$ bubbles per Hubble patch [27], while the single-bubble fit to the DESI Alcock–Paczynski anomaly is independently excluded at 2σ by CMB-dipole and kSZ data [28]. The integer’s signature is the step – fluctuation-level, not background-level – as the ring current’s signature was the integer W in the momentum distribution, not the bath temperature.

4 K3 as a charge lattice: the attractor mechanism is the reading

For type II on $K3 \times T^2$ a dyon is a charge vector $\gamma = (p, q)$ with $p, q \in H^2(K3, \mathbb{Z}) = \Pi_{3,19}$ after U-duality [29, 30]. The attractor equations fix the horizon moduli *from the charges*: $\Omega \propto q - \bar{\tau}p$ with $\tau = (p \cdot q + \sqrt{D})/p^2$, so the attractor K3 has $\text{NS}(S) = \langle p, q \rangle^\perp$ of rank 20 – an attractive K3 whose transcendental lattice is $\langle p, q \rangle$ with Gram matrix $2Q_{p,q}$. The horizon area and the BPS asymptotics read one integer,

$$D(p, q) = (p \cdot q)^2 - p^2 q^2, \quad A/4\pi = \sqrt{-D}, \quad \log \dim \mathcal{H}_{\text{BPS}} = \pi\sqrt{-D} + \dots$$

[31]. The U-duality group acts on charges and preserves D ; but the number of U-inequivalent charges with the same D is the class number $h(D)$ [29]: *many sectors, one entropy, and no symmetry of string theory permuting them.*¹

Theorem 3 (`disc_sl2`, `disc_sl2_invariant`, `disc_electric_magnetic_swap`, `disc_scale`, `disc_zero_of_parallel_area_sl2_invariant`). *For any additive group with a symmetric $\mathbb{Z}$ -bilinear form B and $D(p, q) = B(p, q)^2 - B(p, p)B(q, q)$: $D(ap + bq, cp + dq) = (ad - bc)^2 D(p, q)$; hence D is $\text{SL}(2, \mathbb{Z})$ -invariant, invariant under $(p, q) \mapsto (q, -p)$, scales as t^4 under $(p, q) \mapsto (tp, tq)$, vanishes on (p, kp) , and $\sqrt{-D}$ is constant on orbits.*

¹This is Moore’s own theorem, $N(D) = h(D)$, in the $1/8$ -BPS $K3 \times T^2$ setting stated here; it does not generalise unqualified. For $1/4$ -BPS dyons on heterotic/ T^6 (a different setting), Dabholkar, Gaiotto and Nampuri show the three continuous-duality invariants “do not uniquely specify the state,” and an extra discrete invariant plus the moduli-space chamber (walls of marginal stability [32]) are needed. “Entropy is a function of the sector” is correct as stated here; it is not, unqualified, a claim about every dyon-counting problem in string theory.

Elementary – bilinearity and `ring` – and that is the point: the part of “K3 duality” that carries physics is a polynomial identity on the charge lattice. Three consequences for this programme:

- The K3 that appears is an *output of the sector*. The identification excluded in earlier versions (a superfluid’s dual-length involution as $K3 \times T^2$ mirror symmetry) tried to transfer an *input* geometry; the attractor theorem shows the geometry is determined by charges (p, q) , and a superfluid has no such charges, only $W \in \mathbb{Z}$.
- The Fricke / Dolgachev–Nikulin mirror involution on $\Gamma_0(N) + N$ [33] exchanges lattice polarisations $M \leftrightarrow M^\perp$ inside $\Pi_{3,19}$: a reindexing of the kind of `CompactBoson.partition_dual`, preserving the structure and moving no dyon’s entropy – consistent with `QHFricke`, where the Fricke element fixes an orbit and is not a symmetry of the sectors.
- The cross-domain criterion of [1] gains a clean fourth case: a large duality group, a single integer invariant, physics that reads the invariant – organisation, not cause – with the twist that the level set of the invariant is strictly larger than the orbit. (The Mathieu moonshine of the K3 elliptic genus [34] is a structure on the same lattice; it is not touched here.)

5 What this note does not claim

- Any cosmological number: no boson mass, no Ω , no $w(z)$, no lensing amplitude, no string tension.
- That dark matter is a $U(1)$ field, that Λ sits on a flux lattice, or that dark energy is a top-form flux: these are the hypotheses of the cited papers; the theorems state what follows if they hold.
- Anything about real black holes or about K3 beyond $H^2(K3, \mathbb{Z})$ as a lattice with a form.
- Novelty in the mathematics (bilinearity; a square) or in the physics (Moore; Bousso–Polchinski; Kaloper; Hui et al.; Zhou et al.; Bucher–Spergel are cited, not extended).
- A revival of $\alpha' = \ell_{PC}/H_0$ or of a fluid $\leftrightarrow$ K3 identification; both remain excluded.

Reproducibility

`lean_src/ChargeLattice.lean` (15 theorems; axioms `{propext, Classical.choice, Quot.sound}`); accepted by the Lean kernel and by `nanoda` through `Comparator`; three negative controls – a determinant-2 map, the nucleation step with the wrong sign, Kaloper’s minimum at $N = 0$ – fail); `docs/designs/COSMOLOGY_SECTORS_PROPOSAL.md`; LEDGER CLAIM-047. Every DOI was resolved through Crossref before citation.

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